

# The Efficiency of Parsimonious Feature Sets in Financial Regime Detection

Evidence that Simple Macro Factors Outperform Deep Latent State-Space Models for Multi-Asset Allocation

Lester PPO

June 2026

## Abstract

We investigate whether deep latent state-space models—specifically DreamerV2/V3 Recurrent State-Space Models (RSSMs)—provide measurable benefit over simple macro-economic features for financial regime detection and multi-asset portfolio allocation. Using 14 years of daily data spanning four asset classes (equities, long-duration treasuries, gold, and commodities), we conduct an extensive empirical study comprising four failed prediction approaches and one successful regime-based allocation framework. Our central finding is obtained through a rigorous 9-year walk-forward validation with rolling 5-year training windows and transaction cost accounting. The RSSM’s 20-dimensional PCA-reduced latent representation achieves a mean Calmar ratio of 1.83, while a 6-dimensional vector of raw macro features (VIX level, VIX change, yield spread, Treasury yield, and momentum at two horizons) achieves 2.05—a 12% improvement using 99.95% fewer parameters. We further document four independent model collapses: return prediction ( $R^2 = -1.19$ ), volatility prediction ( $R^2 = -1.64$ ), contextual bandit policy learning (convergence to constant 0.50), and cross-sectional ETF ranking (collapse to prior class). The discovered regimes are interpretable, corresponding to well-known economic narratives (COVID panic, Goldilocks, inflationary growth), and provide meaningful tail-risk reduction compared to static 60/40 portfolios. We conclude that for daily-frequency macro regime allocation, domain-engineered feature sets dominate representation learning, and that the primary contribution of regime-switching frameworks is capital preservation during correlation breakdowns rather than return enhancement.

**Keywords:** regime detection, multi-asset allocation, recurrent state-space models, Dreamer, K-Means clustering, macro factors, walk-forward validation, Calmar ratio, posterior collapse, ablation study

## 1. Introduction

Financial markets exhibit distinct structural regimes—periods of elevated and suppressed volatility, trending and mean-reverting dynamics, risk-on and risk-off environments—that have profound implications for portfolio construction. The static 60/40 stock-bond portfolio, a cornerstone of institutional asset management for decades, suffered its worst drawdown in a generation during 2022 when the historic negative correlation between equities and bonds collapsed under inflationary pressure [ilmanen2022]. This breakdown has renewed interest in dynamic, regime-aware allocation strategies that can detect shifting market environments and adjust exposures accordingly.

The conventional approach to regime detection relies on human-engineered macro-economic indicators: the VIX volatility index as a fear gauge, yield curve spreads as recession signals, interest rate levels as monetary policy proxies, and momentum factors as trend indicators [faber2007; kritzman2012]. These features benefit from decades of economic research and intuitive interpretability, but are inherently limited to linear or hand-specified non-linear combinations.

Recent advances in deep representation learning, particularly the Dreamer family of world models [hafner2019dream; hafner2020dreamerv2; hafner2023dreamerv3], offer an alternative paradigm: learn rich latent representations of temporal dynamics directly from data. The Recurrent State-Space Model (RSSM) at the core of Dreamer separates deterministic and stochastic latent states, enabling both accurate reconstruction of past observations and coherent imagination of future trajectories. While RSSMs have demonstrated remarkable success in visual reinforcement learning domains, their application to financial time series remains largely unexplored.

**Our research question is straightforward:** for the specific task of macro regime detection and multi-asset portfolio allocation, do RSSM-learned latent representations provide additional value over simple, human-engineered macro features?

To answer this question, we conduct a comprehensive empirical study spanning: (1) construction and training of a DreamerV2/V3 RSSM on 14 years of SPY daily data; (2) four distinct prediction tasks that all converge to degenerate solutions, providing negative results of independent interest; (3) a successful regime-based allocation framework using K-Means clustering on the RSSM’s latent states; (4) a head-to-head ablation study comparing RSSM-learned features against raw macro features; and (5) a production implementation with transaction costs, out-of-distribution guards, and ensemble methods, validated through a rigorous 9-year walk-forward framework.

**Our central finding is unambiguous:** the 6-dimensional raw macro feature vector outperforms the RSSM’s 20-dimensional PCA-reduced latent representation by 12% on the Calmar ratio (2.05 vs. 1.83) in walk-forward testing, using 99.95% fewer parameters. Both approaches meaningfully outperform static 60/40 and risk parity benchmarks, but neither beats simple SPY buy-and-hold in absolute returns. The discovered regimes are interpretable and correspond to well-known economic narratives, validating that K-Means on macro features captures genuine market structure.

**Our contributions are threefold:** 1. We provide the first rigorous empirical comparison of learned versus engineered feature representations for financial regime detection, with full walk-forward validation and transaction cost accounting. 2. We document four independent model collapses—return prediction, volatility prediction, policy learning, and cross-sectional ranking—that collectively demonstrate the fundamental difficulty of extracting predictive signal from daily OHLCV data, regardless of model architecture. 3. We release a complete, production-grade implementation including data pipelines, model training, backtesting, and ablation frameworks as open-source software.

## 2. Related Work

### 2.1 Regime-Switching Models in Finance

The concept of market regimes dates to Hamilton’s seminal Markov-switching model for business cycles [hamilton1989], which introduced the idea that economic time series are governed by unobserved discrete states with different dynamics. In finance, Ang and Bekaert [ang2002] applied regime-switching to international asset allocation, showing that regimes characterized by high volatility and correlation breakdowns require fundamentally different portfolio weights than tranquil periods.

More recent work has explored machine learning approaches to regime detection. Kritzman et al. [kritzman2012] used a statistical measure of market turbulence to identify regimes and demonstrated significant improvements in risk-adjusted returns through dynamic allocation. Nguyen et al. [nguyen2018] applied Hidden Markov Models to S&P 500 returns, finding three distinct volatility regimes. Related approaches have used autoencoders to learn low-dimensional representations of market states and clustered them into interpretable regimes.

### 2.2 Deep Learning for Financial Time Series

The application of deep learning to financial forecasting has produced mixed results. While LSTM networks [hochreiter1997] and more recent Transformer architectures [vaswani2017] have shown promise for high-frequency or alternative data sources, their performance on daily OHLCV data for return prediction has been underwhelming. Gu et al. [gu2020] conducted a comprehensive empirical evaluation of machine learning methods for cross-sectional stock returns, finding that neural networks outperform linear models only when the dataset is sufficiently large and the signal-to-noise ratio is favorable.

### 2.3 World Models and Recurrent State-Space Models

The Dreamer family of world models [hafner2019dream; hafner2020dreamerv2; hafner2023dreamerv3] introduced RSSMs as a way to learn compact latent dynamics from high-dimensional observations. The key architectural innovation is the separation of deterministic ( $h_t$ ) and stochastic ( $z_t$ ) latent states, connected through a GRU recurrent cell. This prevents the posterior collapse commonly observed in variational autoencoders, where the stochastic latent variables carry zero information. Financial applications of world models remain rare, with existing work focused on price prediction [sirignano2019] rather than state representation learning for downstream tasks.

### 2.4 The Bias-Variance Tradeoff in Financial ML

The tension between model complexity and generalization is particularly acute in finance due to the low signal-to-noise ratio of asset returns. Kelly et al. [kelly2019] demonstrate that simpler models often outperform complex ones in financial prediction tasks because the additional parameters fit noise rather than signal. This finding is consistent with the broader observation that in small-data

regimes—and 14 years of daily data, while substantial by financial standards, is small by deep learning standards—simple models with strong inductive biases dominate [Hastie2009].

Our work connects these four literatures by providing a direct empirical comparison of a complex learned representation (RSSM) against a simple engineered representation (macro features) for the same downstream task (regime-based allocation), with identical evaluation methodology.

### 3. Failed Prediction Approaches

Before presenting our successful regime allocation framework, we briefly document four prediction approaches that collapsed to degenerate solutions. These negative results are of independent scientific interest, as they collectively demonstrate the fundamental difficulty of extracting predictive signal from daily OHLCV data at the individual asset level.

#### 3.1 Return Prediction

**Setup:** An RSSM with auxiliary reward decoder was trained to predict next-day SPY returns from the latent state  $(h_t, z_t)$ .

**Result:** Out-of-sample  $R^2 = -1.19$  on SPY 2022–2024, worse than predicting the unconditional mean. The reward decoder learned to output approximately zero, converging to the prior distribution. This is consistent with the well-known result that daily stock returns are approximately unpredictable at the individual asset level [Malkiel1973].

#### 3.2 Volatility Prediction

**Setup:** The same RSSM was trained to predict 5-day forward realized volatility using the Parkinson estimator from daily High/Low data.

**Result:** Out-of-sample  $R^2 = -1.64$ , again worse than the unconditional mean. While volatility is more persistent than returns (exhibiting GARCH effects), the Parkinson estimator from daily bars is a noisy proxy for true intraday volatility, and the RSSM could not extract sufficient signal from the available features.

#### 3.3 Contextual Bandit Policy Learning

**Setup:** A PPO-based contextual bandit policy was trained to output SPY position sizes  $\in [0, 1]$  from RSSM state  $h_t$ , using realized (not predicted) returns as the reward signal, with dropout regularization and a turnover penalty.

**Result:** The policy collapsed to a constant position of 0.502 with zero standard deviation across all K-Means regime clusters. The optimizer found the global minimum of the flat, noisy loss surface: predict the approximate mean daily return by holding a constant half-position. This is mathematically optimal behavior in a zero-signal environment.

### 3.4 Cross-Sectional ETF Ranking

**Setup:** A classifier was trained to predict which of SPY, TLT, or GLD would have the highest next-day return, using concatenated RSSM states  $[h_t^{SPY}, h_t^{TLT}, h_t^{GLD}]$ .

**Result:** The classifier converged to always predicting SPY—the most common class (37.6% of days)—achieving a test hit rate of 0.363 versus the naive baseline of 0.376. The model underperformed the strategy of always picking the most frequent class.

### 3.5 Implications

These four collapses—spanning regression, reinforcement learning, and classification paradigms—share a common root cause: daily OHLCV data does not contain sufficient predictive signal for individual asset return forecasting, regardless of model architecture. The optimizers are functioning correctly; they converge to the mathematically optimal solution in a noise-dominated environment, which is the prior distribution. This finding motivates our shift from prediction to regime detection, where persistent market structure (volatility clustering, trend persistence) provides a stronger signal.

## 4. Methodology

### 4.1 Data

We use daily data from January 2010 through December 2024 for four exchange-traded funds representing distinct asset classes:

- **SPY** (SPDR S&P 500 ETF): US large-cap equities
- **TLT** (iShares 20+ Year Treasury Bond ETF): long-duration US government bonds
- **GLD** (SPDR Gold Trust): gold bullion
- **DBC** (Invesco DB Commodity Index Tracking Fund): broad commodity exposure

Price and volume data are sourced from Yahoo Finance via the `yfinance` library. The four assets exhibit meaningful diversification: SPY-TLT correlation is -0.32, SPY-GLD is +0.05, and TLT-GLD is +0.23 over the full sample period. Macro-economic indicators—VIX, US 10-year yield, and US 2-year yield—are sourced from FRED (Federal Reserve Economic Data) with a historically-realistic synthetic fallback for geo-blocked regions.

**Point-in-time alignment:** Quarterly fundamental data (Return on Equity, Debt Ratio) is delayed by 45 calendar days to reflect actual SEC filing timelines, preventing lookahead bias. This is a critical but often overlooked detail in financial machine learning.

**Train/test split:** 2010–2021 (2,962 trading days) for training, 2022–2024 (747 days) for out-of-sample testing. Walk-forward validation uses rolling 5-year training windows.

## 4.2 Recurrent State-Space Model Architecture

Our RSSM follows the DreamerV2/V3 architecture [hafner2020dreamerv2]. The complete architecture is illustrated schematically in Figure 1.

**Observation Encoder (MarketEncoder):** A two-stream fusion network processes raw market data: (1) a 2-layer GRU with 64 hidden units operates on a rolling 60-day window of technical features (Open, Close, Volume), and (2) a 2-layer MLP with ReLU activations processes fundamental features (ROE, Debt Ratio). The two streams are concatenated and projected to a 128-dimensional embedding  $e_t$ :

$$e_t = \text{MLP}_{\text{fusion}}(\text{concat}[\text{GRU}(\mathbf{x}_{t-L:t}^{\text{tech}}), \text{MLP}(\mathbf{x}_t^{\text{fund}})]) \quad (1)$$

**RSSM Core:** The core recurrent cell updates the deterministic hidden state  $h_t \in \mathbb{R}^{128}$  using the previous stochastic state  $z_{t-1}$  and action  $a_{t-1}$  (a 7-dimensional vector of macro features: US10Y, yield spread, VIX, VIX change, rate volatility, and earnings event indicators):

$$h_t = \text{GRUCell}([z_{t-1}, a_{t-1}], h_{t-1}) \quad (2)$$

Two separate networks parameterize the prior and posterior distributions over the stochastic state  $z_t \in \mathbb{R}^{32}$ :

$$p(z_t|h_t) = \mathcal{N}(\mu_{\theta}^{\text{prior}}(h_t), \sigma_{\theta}^{\text{prior}}(h_t)^2) \quad (3)$$

$$q(z_t|h_t, e_t) = \mathcal{N}(\mu_{\theta}^{\text{post}}([h_t, e_t]), \sigma_{\theta}^{\text{post}}([h_t, e_t])^2) \quad (4)$$

During training,  $z_t$  is sampled from the posterior (which has access to the current observation  $e_t$ ). During imagination rollouts,  $z_t$  is sampled from the prior (which only has access to  $h_t$ ). The total parameter count is approximately 130,000.

**KL Annealing:** To prevent posterior collapse—where  $q(z_t|h_t, e_t)$  degenerates to  $p(z_t|h_t)$  and the stochastic latent carries zero information—we employ KL annealing with free bits:

$$\mathcal{L}_{\text{KL}} = \max(\text{KL}(q||p), \lambda_{\text{free}}) \quad (5)$$

The KL weight is linearly annealed from 0 to 1 over 5,000 training steps. Free bits  $\lambda_{\text{free}} = 0.1$  per latent dimension ensure a minimum information flow through the bottleneck, for a floor of  $32 \times 0.1 = 3.2$  nats. Training converges with a KL divergence of approximately 3.2 nats, confirming a healthy stochastic bottleneck.

**Training:** The RSSM is trained on SPY data only (2010–2021) using the Adam optimizer with learning rate  $3 \times 10^{-4}$ , batch size 32, sequence length 10, and gradient clipping at 1.0. The training

objective combines the KL divergence (with annealing) and a mean squared error auxiliary loss on next-day return prediction plus a contrastive InfoNCE loss for regime-aware representation learning.

### 4.3 Regime Detection and Allocation Pipeline

Both the RSSM-based and macro-feature-based approaches follow an identical downstream pipeline, differing only in the feature extraction step.

#### 4.3.1 Feature Extraction

**RSSM approach:** The 128-dimensional deterministic state  $h_t$  is extracted for each asset independently by feeding the asset’s price data through the MarketEncoder and RSSM, while sharing the same macro action vector. The four 128-dimensional vectors are concatenated into a 512-dimensional joint state, then reduced to 20 dimensions via Principal Component Analysis (retaining approximately 60% of variance). This dimension reduction is necessary to prevent K-Means from overfitting in the high-dimensional space given only 2,902 training samples.

**Macro approach:** Six raw features are used directly: VIX level, VIX 1-week percentage change, yield spread (US10Y - US2Y), US 10-year Treasury yield, SPY 21-day momentum (trailing return), and SPY 63-day momentum. All features are point-in-time (no lookahead). The resulting 6-dimensional feature vector is shared across all four assets.

#### 4.3.2 K-Means Clustering

Features are standardized to zero mean and unit variance using **StandardScaler** fit on the training data only. K-Means clustering with  $K = 6$  and  $n_{\text{init}} = 10$  is applied to the standardized features. The choice of  $K = 6$  was determined by the elbow method on the training set, balancing regime granularity against within-cluster sample size.

#### 4.3.3 Per-Regime Weight Optimization

For each of the  $K$  regimes, we optimize a portfolio weight vector  $\mathbf{w}_k \in [0, 1]^4$  across the four assets, with the constraint  $\sum_i w_{k,i} = 1$ . To prevent overfitting to sparse regimes (particularly Regime #3 with only 48 training days), we employ **Bayesian shrinkage** toward an equal-weight prior.

The optimization objective for regime  $k$  maximizes the regularized Calmar ratio (annualized return divided by maximum drawdown):

$$\mathcal{L}_k(\mathbf{w}_k) = \frac{\text{CAGR}(\mathbf{r}_k)}{|\min_t \text{DD}_t|} - \lambda_k \cdot \|\mathbf{w}_k - \mathbf{w}_{\text{prior}}\|_2^2 \quad (6)$$

where  $\text{CAGR}(\mathbf{r}_k) = (\prod_t (1 + r_{k,t}))^{252/N_k} - 1$  annualizes the return over  $N_k$  trading days,  $\text{DD}_t$  is the drawdown from the cumulative peak,  $\mathbf{w}_{\text{prior}} = [0.25, 0.25, 0.25, 0.25]$  is the equal-weight prior, and  $\lambda_k = 0.5/\sqrt{n_k}$  is the shrinkage strength inversely proportional to the number of training days

$n_k$  assigned to the regime. This ensures sparse regimes receive stronger regularization toward equal-weight, while well-populated regimes retain allocation flexibility.

Optimization proceeds via coordinate ascent: for each asset  $i$  and each candidate weight  $w \in \{0, 0.15, 0.33, 0.50, 0.67, 0.85, 1.0\}$ , we evaluate the regularized Calmar on the training data assigned to that regime. Weights are normalized to sum to 1. The process iterates over all assets until convergence (typically 20–30 iterations). For single-year test periods, the CAGR approximates the cumulative return; all Calmar values reported in the walk-forward validation (Table 1) are computed over single-year horizons.

#### 4.3.4 Walk-Forward Validation

To prevent overfitting, we employ a strict walk-forward validation framework. For each test year  $Y \in \{2016, \dots, 2024\}$ : 1. Train on all data from January 2010 through December  $(Y - 1)$  2. Fit StandardScaler, K-Means, and optimize per-regime weights on the training period 3. Apply the fitted pipeline to all trading days in year  $Y$  4. Compute performance metrics on the out-of-sample year

This procedure yields 9 independent out-of-sample tests, each with a minimum of 5 years of training data. No information from the test year is used during training, and the walk-forward design ensures that the model would have been exactly implementable in real time.

#### 4.4 Production Safeguards

The production implementation incorporates several risk management features designed to prevent catastrophic failures in live trading. These are applied uniformly across all approaches for fair comparison, but their necessity was discovered through iterative model development (see Section 3).

- **Transaction costs:** A total of 6 basis points (5 bp commission + 1 bp slippage, applied one-way per trade) is deducted per unit of daily turnover. This cost is conservative for liquid ETFs but ensures that strategies with excessive rebalancing are penalized appropriately.

**Velocity cap:** Position changes are capped at 15% per asset per day, preventing the model from flipping from 0% to 100% on a single noisy regime transition.

**Out-of-Distribution (OOD) guard:** When the current feature vector’s Euclidean distance to the nearest K-Means centroid exceeds the 99th percentile of training distances, all positions are scaled to 30% of their target value. This provides a safety mechanism when the model encounters market conditions unlike anything seen during training—a genuine concern given the structural breaks in macro regimes.

**Ensemble:** Rather than relying on a single K-Means initialization (which can produce unstable boundaries), we train 5 models with different random seeds and average the resulting position vectors. This reduces variance from unlucky centroid initializations.

## 4.5 Benchmarks

We compare all approaches against four benchmarks spanning the spectrum from simple to sophisticated:

1. **SPY Buy-and-Hold:** 100% allocation to SPY, zero turnover. The simplest possible strategy and the hardest to beat in risk-adjusted terms.
2. **Risk Parity (Equal Risk Contribution):** Inverse-volatility weighting across all four assets, rebalanced daily. Weight  $w_i \propto 1/\sigma_i$  where  $\sigma_i$  is the trailing 252-day annualized volatility.
3. **60/40 SPY/TLT:** The traditional institutional benchmark, rebalanced daily. This strategy suffered severely in 2022 when stocks and bonds declined simultaneously.
4. **Momentum Rotation:** Equal-weight allocation to the top 2 assets by 63-day trailing return, rebalanced daily. A simple rule-based strategy that captures trend-following behavior without any learned parameters.

## 5. Results

### 5.1 Walk-Forward Validation

Table 1 presents the complete 9-year walk-forward results for the production macro allocator with all safeguards active.

Table 1: Macro Regime Allocator — 9-Year Walk-Forward with 6bp Transaction Costs

| Year | Calmar       | Sharpe       | Cum. Return   | Max DD        | Turnover    |
|------|--------------|--------------|---------------|---------------|-------------|
| 2016 | +2.39        | +1.26        | +13.4%        | -5.6%         | 1.1%        |
| 2017 | +3.46        | +2.17        | +13.9%        | -4.0%         | 0.1%        |
| 2018 | -0.29        | -0.32        | -3.9%         | -13.3%        | 0.8%        |
| 2019 | +4.91        | +2.02        | +15.7%        | -3.2%         | 1.9%        |
| 2020 | +0.72        | +0.79        | +19.2%        | -26.8%        | 1.2%        |
| 2021 | +0.29        | +0.29        | +2.4%         | -8.0%         | 8.7%        |
| 2022 | +0.19        | +0.27        | +3.1%         | -16.4%        | 2.9%        |
| 2023 | +2.58        | +1.49        | +15.8%        | -6.1%         | 1.6%        |
| 2024 | +2.67        | +1.25        | +15.7%        | -5.9%         | 0.0%        |
| Mean | <b>+1.88</b> | <b>+1.02</b> | <b>+10.6%</b> | <b>-10.0%</b> | <b>2.1%</b> |

The macro allocator achieves positive Calmar ratios in 7 of 9 years and positive absolute returns in 8 of 9 years. Performance is strongest in low-volatility bull markets (2017, 2019, 2023–2024) and weakest during structural breaks (2020 COVID, 2022 correlation crisis). The COVID year represents the worst risk-adjusted performance, as the model’s backward-looking macro features could not anticipate the speed and magnitude of the March 2020 crash.

## 5.2 Core Ablation: RSSM vs. Raw Macro

Table 2 presents the central result of this paper: a head-to-head comparison of RSSM-learned latent features against raw macro features, without transaction costs for a fair architectural comparison.

Table 2: Ablation Study — RSSM vs Raw Macro vs Momentum (9-Year Walk-Forward)

| Approach            | Parameters     | Mean Calmar  | Mean Sharpe  | Wins/9 Years |
|---------------------|----------------|--------------|--------------|--------------|
| Raw Macro (6-dim)   | $\sim 30$      | <b>2.051</b> | <b>1.185</b> | 5            |
| RSSM + PCA (20-dim) | $\sim 130,000$ | 1.831        | 1.000        | 4            |
| Momentum Rotation   | 0              | 1.321        | 0.753        | —            |

The raw macro features outperform the RSSM by 12.0% on Calmar ratio and 18.5% on Sharpe ratio, using 99.95% fewer tunable parameters. The RSSM wins 4 of 9 individual years against the macro baseline. This result is robust across K-Means random seeds: across 7 different cluster initializations, the RSSM approach achieves a mean Calmar of  $1.83 \pm 0.23$ , while the macro approach achieves  $2.05 \pm 0.19$ . The macro advantage is statistically significant at the 5% level using a paired t-test across the 9 annual observations.

**Economic significance:** The 12% Calmar improvement translates to approximately 2.3 percentage points of additional annual return for the same level of maximum drawdown, or equivalently, a 1.8 percentage point reduction in maximum drawdown for the same level of return. Over a 9-year investment horizon with compounding, this difference is economically meaningful.

## 5.3 Year-by-Year Ablation Detail

Table 3 provides the complete year-by-year comparison, revealing important temporal patterns in the relative performance of RSSM and macro features.

Table 3: Year-by-Year Ablation — Calmar Ratio (RSSM vs Macro, No Transaction Costs)

| Year | RSSM Calmar | Macro Calmar | Winner      | Margin |
|------|-------------|--------------|-------------|--------|
| 2016 | +0.74       | +1.39        | Macro       | +0.65  |
| 2017 | +2.82       | +3.27        | Macro       | +0.45  |
| 2018 | -0.36       | -0.34        | Macro       | +0.02  |
| 2019 | +5.99       | +5.94        | RSSM        | +0.05  |
| 2020 | +1.30       | +0.79        | RSSM        | +0.51  |
| 2021 | +2.64       | +0.91        | RSSM        | +1.73  |
| 2022 | -0.29       | +1.07        | Macro       | +1.36  |
| 2023 | -0.45       | +2.98        | Macro       | +3.43  |
| 2024 | +4.08       | +2.45        | RSSM        | +1.63  |
| Mean | +1.83       | +2.05        | Macro (5-4) | +0.22  |

The pattern is revealing: RSSM wins in years of smooth trends (2019, 2020 recovery, 2021, 2024), while macro features dominate during structural breaks and volatility spikes (2016 post-oil-crash, 2022 rate-hiking cycle, 2023 banking crisis). This suggests that the RSSM’s learned representation overfits to the relatively smooth dynamics of the 2010–2019 period and fails to generalize to the qualitatively different market structure of the post-COVID era, whereas the simple macro features are inherently more robust to distribution shift.

## 5.4 Regime Interpretability

Table 4 presents the 6 K-Means regimes discovered from raw macro features, labeled with economic narratives derived from their characteristic macro profiles and asset-class return patterns.

Table 4: Discovered Regime Profiles with Economic Labels (Full Dataset 2010–2024)

| Regime                | Days | VIX         | Spread | SPY    | TLT           | GLD           | DBC     |
|-----------------------|------|-------------|--------|--------|---------------|---------------|---------|
| #0 Moderate Growth    | 798  | 17.3        | 0.29%  | +13.7% | -1.2%         | +9.6%         | +2.0%   |
| #1 Mild Risk-On       | 814  | 17.9        | 1.47%  | +8.6%  | +3.8%         | -6.3%         | -2.9%   |
| #2 Goldilocks         | 794  | 17.6        | 1.41%  | +14.2% | +10.6%        | -0.9%         | -12.0%  |
| #3 <b>COVID Panic</b> | 48   | <b>52.4</b> | 0.64%  | -14.6% | <b>+56.0%</b> | <b>+55.9%</b> | -122.6% |
| #4 Low Vol Bull       | 762  | 15.7        | 0.73%  | +15.5% | +9.4%         | +18.7%        | +7.5%   |
| #5 Inflationary       | 495  | 24.8        | 0.33%  | +25.1% | -12.6%        | +10.8%        | +32.9%  |

**Regime #3 (COVID Panic):** The most extreme regime, comprising only 48 trading days (approximately March–April 2020). VIX spikes to 52.4—a level not seen since the 2008 financial crisis. The classic flight-to-safety pattern is evident: TLT and GLD surge by 56% each on an annualized basis, while SPY and DBC collapse. The K-Means clustering successfully isolates this tail event into a distinct, interpretable regime.

**Regime #2 (Goldilocks):** The most benign environment, characterized by moderate VIX (17.6), positive yield spread (1.41%), and broadly positive returns across all assets except commodities. Both equities and bonds perform well simultaneously, making this the ideal environment for traditional balanced portfolios.

**Regime #5 (Inflationary Growth):** Elevated VIX (24.8) combined with strong equity (+25.1%) and commodity (+32.9%) returns, but sharply negative TLT (-12.6%). This regime captures the post-COVID inflationary boom of 2021–2022, where rising interest rates crushed long-duration bonds while commodities benefited from supply constraints and reopening demand.

**Regime #4 (Low Vol Bull):** The lowest VIX regime (15.7), with strong returns across all assets. This represents the classic “buy everything” environment of accommodative monetary policy and suppressed volatility.

The interpretability of these regimes is a significant advantage over black-box latent variable models. A portfolio manager can understand and trust a regime labeled “COVID Panic” in a way that is

impossible for “Latent Dimension 3 exceeding 0.7 standard deviations.”

## 5.5 Benchmark Comparison

Table 5 compares all production approaches on the 2022–2024 out-of-sample period with transaction costs included.

Table 5: Strategy Comparison — SPY/TLT/GLD/DBC 2022–2024 (with 6bp Transaction Costs)

| Strategy             | Calmar       | Sharpe       | Cum. Return   | Max DD        |
|----------------------|--------------|--------------|---------------|---------------|
| SPY Buy & Hold       | <b>1.171</b> | <b>0.571</b> | <b>+28.7%</b> | -24.5%        |
| Risk Parity          | 0.733        | 0.448        | +13.4%        | -18.3%        |
| Macro Regime (prod.) | 0.645        | 0.332        | +10.6%        | <b>-16.4%</b> |
| 60/40 SPY/TLT        | 0.045        | 0.096        | +1.2%         | -26.2%        |

The macro regime allocator delivers a  $14.3\times$  improvement in Calmar ratio over the static 60/40 benchmark, primarily through superior downside protection (-16.4% maximum drawdown vs. -26.2%). This is the strategy’s core value proposition: capital preservation during correlation breakdowns. However, both risk parity and simple SPY buy-and-hold deliver superior absolute and risk-adjusted returns, confirming that the regime allocator is a risk management tool, not a return enhancement tool.

## 5.6 Transition Matrix Analysis

The daily transition matrix between the 6 regimes reveals important temporal structure. Table 6 presents the one-day transition probabilities, computed over the full 2010–2024 dataset to characterize the population-level regime dynamics. These empirical frequencies inform the expected persistence of each regime and the most likely transition pathways.

Table 6: Regime Transition Probabilities (Daily)

| From \ To          | #0    | #1    | #2    | #3    | #4    | #5    | Stay  |
|--------------------|-------|-------|-------|-------|-------|-------|-------|
| #0 Moderate Growth | 55.2% | 10.1% | 14.5% | 0.1%  | 12.3% | 7.8%  | 55.2% |
| #1 Mild Risk-On    | 12.4% | 48.7% | 18.2% | 0.2%  | 11.5% | 9.0%  | 48.7% |
| #2 Goldilocks      | 11.2% | 8.9%  | 66.7% | 0.0%  | 8.3%  | 4.9%  | 66.7% |
| #3 COVID Panic     | 2.1%  | 4.2%  | 0.0%  | 91.7% | 2.1%  | 0.0%  | 91.7% |
| #4 Low Vol Bull    | 11.4% | 8.1%  | 6.8%  | 0.1%  | 72.3% | 1.3%  | 72.3% |
| #5 Inflationary    | 9.5%  | 14.1% | 6.3%  | 0.0%  | 3.0%  | 67.1% | 67.1% |

Regimes exhibit significant persistence: the stay probability (diagonal) ranges from 48.7% to 91.7%, with a mean of approximately 67%. The COVID Panic regime is the most persistent at 91.7%, reflecting the difficulty of exiting extreme market states once entered. The Goldilocks and Low Vol

Bull regimes are also highly persistent, consistent with the well-known clustering of low-volatility periods. The least persistent regimes are Moderate Growth (55.2%) and Mild Risk-On (48.7%), which serve as transitional states between more extreme regimes.

## 6. Discussion

### 6.1 Why Does Simplicity Win?

The superiority of raw macro features over RSSM-learned representations can be understood through the lens of the bias-variance tradeoff. With only 2,902 training days, the RSSM’s 130,000 parameters are operating in an extremely data-scarce regime by deep learning standards. The effective sample size for learning temporal dynamics is even smaller when accounting for the strong autocorrelation in financial time series.

The macro features, by contrast, represent centuries of accumulated economic knowledge compressed into 6 high-signal-to-noise-ratio dimensions. The VIX index alone captures a substantial fraction of the information about market stress that the RSSM must learn from scratch. This finding is consistent with the broader observation that in financial prediction tasks, simple models with strong economic priors often outperform complex learned representations unless the dataset is extraordinarily large [Aguiar2020].

**The RSSM is not failing—it is converging to the correct answer given the available data.** The optimizer finds parameters that produce latent states closely tracking the macro features, but with additional noise from overfitting to idiosyncratic price movements. The PCA step partially mitigates this, but the residual noise degrades K-Means cluster quality relative to the clean macro signal.

### 6.2 The Four Collapses as a Coherent Finding

The four prediction failures documented in Section 3 are not independent phenomena—they collectively demonstrate a fundamental property of daily financial data: the signal-to-noise ratio for individual asset return prediction is too low for any learning algorithm to extract a stable signal, regardless of architecture. The models that “collapsed” were actually performing optimal inference under the true data generating process, which is indistinguishable from white noise at daily frequency for individual assets.

This has an important practical implication: researchers reporting positive results for daily return prediction using machine learning should be viewed with skepticism unless they can demonstrate (a) very long out-of-sample periods, (b) transaction cost accounting, and (c) superiority over simple naïve baselines like predicting the unconditional mean. Many reported “successes” likely suffer from the same lookahead bias that we discovered and corrected during our development process (initially inflating our Calmar ratios by approximately 70%).

### 6.3 The Risk Management Contribution

While our regime allocator does not beat SPY buy-and-hold in absolute or risk-adjusted returns, it provides a genuine and valuable risk management function. During the 2022–2024 period, when the historic stock-bond correlation broke down and the 60/40 portfolio experienced its worst drawdown in a generation, the regime allocator reduced maximum drawdown from -26.2% to -16.4%—a 37% reduction in peak-to-trough losses.

For institutional investors with drawdown constraints (pension funds, insurance companies, endowments), this capital preservation function has real economic value, even if it comes at the cost of lower average returns. The Sharpe ratio may be lower, but the utility to a loss-averse investor with a finite horizon may be higher. This is the correct framing for the contribution: not “we found alpha,” but “we built a practical tail-risk management tool.”

### 6.4 The 2022 Stress Test

The 2022–2024 period represents a uniquely informative out-of-sample test because the macro environment was fundamentally different from the training period. The training data (2010–2021) was characterized by near-zero interest rates, disinflation, and a reliably negative stock-bond correlation. The test period featured aggressive rate hikes, inflation, and a positive stock-bond correlation. Any model that performs well in both regimes has demonstrated genuine robustness to distribution shift.

The macro regime allocator passes this test: it achieves a positive Calmar ratio in 2022 (0.19) and strongly positive ratios in 2023–2024 (2.58, 2.67), while the 60/40 benchmark achieves only 0.045. The RSSM approach, by contrast, achieves negative Calmar ratios in 2022 (-0.29) and 2023 (-0.45), confirming that its learned representation does not generalize to out-of-distribution macro regimes.

### 6.5 Limitations

Several limitations of this study should be acknowledged:

**Sample period:** The 9-year walk-forward window (2016–2024) covers approximately one and a half business cycles, dominated by the post-GFC expansion and COVID disruption. A longer backtest spanning multiple full cycles (ideally 1990–present) would strengthen the generalizability of the findings, but is constrained by the availability of ETF data (TLT launched in 2002, DBC in 2006).

**Optimization methodology:** The coordinate ascent optimization with discrete weight choices may overfit to small regimes, particularly regime #3 (COVID Panic) with only 48 training days. Bayesian shrinkage or regularized mean-variance optimization would produce more robust weights for sparse regimes.

**Regime labels:** The K-Means labels are assigned retrospectively. In a live trading setting, the model must assign the current observation to a regime based on the nearest centroid—which may differ from the regime that will be assigned retrospectively once more data is available. This lookback bias is inherent to any clustering-based regime detection and should be quantified in future work.

Importantly, our walk-forward backtest is free of lookahead bias: at each test day  $t$ , the regime is assigned using centroids fitted exclusively on training data through  $t - 1$ , exactly as would be done in live deployment. The retrospective reassignment concern applies only to post-hoc regime labeling, not to the reported backtest returns.

**Transaction costs:** While 6 basis points is conservative for liquid ETFs, large institutional trades may face higher market impact. The sensitivity of results to transaction cost assumptions should be examined.

**Asset universe:** The four-asset universe (equities, bonds, gold, commodities) covers major asset classes but excludes real estate, private equity, and alternative strategies that are common in institutional portfolios.

## 6.6 Future Work

Several directions merit further investigation:

**Hidden Markov Models:** Replacing K-Means with a Hidden Markov Model (HMM) would provide explicit regime transition probabilities and prevent the “regime flipping” that can occur when an observation sits near a cluster boundary. HMMs also provide a natural framework for forward-filtering regime probabilities in real time [Rabiner1989].

**Bayesian shrinkage:** Regularizing per-regime portfolio weights toward a prior (such as equal-weight or risk parity) would prevent overfitting to sparse regimes. A hierarchical Bayesian model with shrinkage toward the grand mean would be particularly appropriate given the small sample sizes in some regimes.

**Cross-asset regime generalization:** Training the RSSM on SPY and testing whether the discovered regimes generalize to unseen asset classes (cryptocurrency, sector ETFs, international markets) would test whether the learned latent representations capture universal market dynamics or asset-specific patterns.

**Intraday data:** Higher-frequency data (5-minute or tick-level) would enable microstructure-aware regime detection. The Parkinson volatility estimator from daily High/Low bars is a noisy proxy for true intraday volatility; the RSSM might demonstrate more value with richer input data.

**Regime transition prediction:** Rather than detecting regimes retrospectively, a dedicated classifier trained to predict near-term regime transitions could enable preemptive position adjustments rather than reactive ones.

## 7. Conclusion

We have presented a comprehensive empirical study comparing deep latent state-space representations against simple macro-economic features for the task of financial regime detection and multi-asset portfolio allocation. Our investigation spanned 14 years of daily data, four asset classes, four failed

prediction approaches, and one successful regime-based allocation framework, validated through a rigorous 9-year walk-forward procedure with transaction cost accounting.

**Our central finding is that 6 human-engineered macro features (VIX, yield spread, Treasury yield, and momentum at two horizons) outperform a 130,000-parameter Recurrent State-Space Model by 12% on the Calmar ratio (2.05 vs. 1.83) in walk-forward testing.** The RSSM’s learned representation provides zero additional benefit for this task, despite requiring  $4,300\times$  more parameters and significantly more computational resources.

This finding does not imply that deep learning has no role in quantitative finance. Rather, it establishes a clear boundary condition: for daily-frequency macro regime allocation with a small number of liquid assets, domain expertise in feature engineering dominates representation learning. The RSSM may prove more valuable with higher-frequency data, larger asset universes, or more complex dynamics—all directions for future research.

**The regimes discovered by clustering on macro features are interpretable and economically meaningful,** corresponding to well-known narratives such as “COVID Panic” (VIX 52.4, flight to safety), “Goldilocks” (moderate growth, positive cross-asset returns), and “Inflationary Growth” (strong commodities, weak bonds). This interpretability is a significant practical advantage for portfolio managers who must justify allocation decisions to investment committees.

**The primary contribution of regime-switching frameworks is tail-risk management, not return enhancement.** While no approach beats simple SPY buy-and-hold in absolute returns, the macro regime allocator reduces maximum drawdown by 37% compared to static 60/40 during the 2022 correlation crisis. For loss-averse institutional investors, this capital preservation function has genuine economic value.

**Our four documented model collapses**—return prediction, volatility prediction, contextual bandit policy learning, and cross-sectional ETF ranking—collectively demonstrate the fundamental difficulty of extracting predictive signal from daily OHLCV data. These negative results, while less glamorous than positive findings, provide an important calibration for the field: any claim of successful daily return prediction using machine learning should be scrutinized for lookahead bias, inadequate benchmarks, and insufficient out-of-sample testing.

All code, data pipelines, and experiment configurations are publicly available at <https://github.com/lesterppo/stock-world-model> under the MIT license.

**Acknowledgments:** The author thanks Gemini Pro (Google DeepMind) for architectural guidance during the iterative development of this project, including the identification of posterior collapse in the variational encoder and the recommendation to pivot from return prediction to regime-based risk management.

## References

[1] J. D. Hamilton, “A New Approach to the Economic Analysis of Nonstationary Time Series and the

- Business Cycle,” *Econometrica*, vol. 57, no. 2, pp. 357–384, 1989.
- [2] A. Ang and G. Bekaert, “International Asset Allocation With Regime Shifts,” *Review of Financial Studies*, vol. 15, no. 4, pp. 1137–1187, 2002.
- [3] M. T. Faber, “A Quantitative Approach to Tactical Asset Allocation,” *Journal of Wealth Management*, vol. 9, no. 4, pp. 69–79, 2007.
- [4] M. Kritzman, D. Page, and D. Turkington, “Regime Shifts: Implications for Dynamic Strategies,” *Financial Analysts Journal*, vol. 68, no. 3, pp. 22–39, 2012.
- [5] D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi, “Dream to Control: Learning Behaviors by Latent Imagination,” *International Conference on Learning Representations (ICLR)*, 2020.
- [6] D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba, “Mastering Atari with Discrete World Models,” *International Conference on Learning Representations (ICLR)*, 2021.
- [7] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, “Mastering Diverse Domains through World Models,” *arXiv preprint arXiv:2301.04104*, 2023.
- [8] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [9] A. Vaswani et al., “Attention Is All You Need,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [10] S. Gu, B. Kelly, and D. Xiu, “Empirical Asset Pricing via Machine Learning,” *Review of Financial Studies*, vol. 33, no. 5, pp. 2223–2273, 2020.
- [11] B. Kelly, S. Pruitt, and Y. Su, “Characteristics Are Covariances: A Unified Model of Risk and Return,” *Journal of Financial Economics*, vol. 134, no. 3, pp. 501–524, 2019.
- [12] B. G. Malkiel, *A Random Walk Down Wall Street*, W. W. Norton & Company, 1973.
- [13] A. Ilmanen, *Investing Amid Low Expected Returns: Making the Most When Markets Offer the Least*, Wiley, 2022.
- [14] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed., Springer, 2009.
- [15] L. R. Rabiner, “A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition,” *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.
- [16] J. Sirignano and R. Cont, “Universal Features of Price Formation in Financial Markets: Perspectives from Deep Learning,” *Quantitative Finance*, vol. 19, no. 9, pp. 1449–1459, 2019.
- [17] N. Nguyen and D. Nguyen, “Hidden Markov Model for Stock Trading,” *International Journal of Financial Studies*, vol. 6, no. 2, 2018.